

iStocks: AI-Powered Stock Analysis Platform

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1 Introduction

iStocks is a cutting-edge financial platform that combines real-time stock data with advanced Artificial Intelligence to provide actionable insights. Built for both novice traders and experienced investors, iStocks offers a suite of tools including AI-driven analysis, natural language database querying, and paper trading capabilities.

2 Key Features

2.1 1. AI Stock Analysis

Leveraging Google’s **Gemini 2.5 Pro** model, iStocks analyzes stock performance, technical indicators (RSI, MACD, SMA), and market sentiment to generate comprehensive

reports.

- **Real-time Insights:** Get up-to-the-minute analysis on any stock.
- **Technical & Fundamental Breakdowns:** Understand the *why* behind the price movements.
- **Buy/Sell/Hold Recommendations:** AI-generated ratings based on multiple data points.

2.2 2. Natural Language Database Chat

Interact with our extensive financial database using plain English. No SQL knowledge required.

- **Ask Questions:** "Show me companies with PE ratio less than 20 and revenue growth over 15%."
- **Comparisons:** "Compare HDFCBANK and ICICIBANK profitability."
- **Context Aware:** The AI remembers your conversation history for follow-up questions.

2.3 3. Real-Time Data Streaming

Access live stock prices via our high-performance WebSocket architecture hosted on AWS EC2.

- **Low Latency:** Millisecond-level updates for tracked stocks.
- **Market Breadth:** Covers major indices (NIFTY 50) and top traded stocks.

2.4 4. Sangraha (Collections)

Curated lists of stocks based on specific themes or strategies.

- **Theme-based investing:** Find stocks in sectors like EV, Green Energy, AI, etc.
- **Expert Picks:** Collections managed by financial experts.

3 Getting Started

3.1 Prerequisites

- Node.js (v18 or higher)
- Python (v3.10 or higher)
- PostgreSQL Database
- API Keys: Angel One (for data), Google Gemini (for AI)

3.2 Installation

1. Clone the Repository

```
1 git clone https://github.com/yourusername/istocks-p.git
2 cd stocks-p
```

2. Install Dependencies

```
1 # Frontend
2 npm install
3
4 # Backend Scripts
5 python3 -m venv venv
6 source venv/bin/activate
7 pip install -r requirements.txt
```

3. Configure Environment Create a '.env' file in the root directory:

```
1 DATABASE_URL="postgresql://user:password@host:5432/dbname"
2 GEMINI_API_KEY="your_api_key"
3 ANGELONE_API_KEY="your_angel_key"
```

4. Run the Application

```
1 npm run dev
```

Access the platform at <http://localhost:3000>.

4 Usage Guide

4.1 analyzing a Stock

Navigate to the dashboard and search for a stock symbol (e.g., "RELIANCE"). Click on the stock card to view the detailed analysis page. The AI report will generate automatically.

4.2 Using Database Chat

Go to the "Database Chat" page. Type your query in the input box. For example:

"List top 5 IT companies by market cap."

The system will generate the SQL query, fetch the data, and present it in a readable table or chart.

5 Technical Architecture

- **Frontend:** Next.js 14 (App Router), Tailwind CSS, Framer Motion
- **Backend:** Next.js API Routes, Python (Data Fetching)
- **Database:** PostgreSQL (Azure) with Prisma ORM
- **Live Server:** Python FastAPI + WebSockets (hosted on AWS EC2)
- **AI Engine:** Google Gemini 2.5 Pro via Vercel AI SDK

6 License

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